

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No .....

## **EC-7005(3)-CBGS**

### **B.E. VII Semester**

Examination, June 2020

## **Choice Based Grading System (CBGS)**

### **Operating Systems**

*Time : Three Hours*

*Maximum Marks : 70*

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.  
iii) Assume suitable data wherever necessary.

1. a) Define the term operating system? Discuss the various services provided by an operating system? 7  
b) Differentiate between:-
  - i) Buffering and Spooling
  - ii) System Calls and System Boot 7
2. a) Discuss Multiprogramming verses single user systems in terms of throughput and CPU utilization. 7  
b) Discuss various file allocation and access methods. Compare their advantages and disadvantages. 7
3. a) Explain various free space management techniques? Give their advantages and disadvantages also. 7  
b) Discuss structure of regular files. How Unix system maintain Directories? 7
4. a) What is Scheduling? Explain short, medium and long term scheduler? 7  
b) Define deadlock. Explain how it causes issues for the system for resource sharing. 7

EC-7005(3)-CBGS

PTO

[2]

5. a) What is Semaphore? How can we achieve the synchronization using it? 7  
b) Explain process management in Linux. 7
6. a) Describe the role of Virtual memory. Explain its mapping through MMU. 7  
b) What is meant by Thrashing? Explain various causes of thrashing. 7
7. a) Explain FIFO and LRU page replacement algorithm in detail. 7  
b) What is Distributed Operating System? Discuss its types along with design issues. 7
8. a) Discuss various security threats in file system of OS. 7  
b) Write short notes:- 7
  - i) Distributed Shared Memory (DSM)
  - ii) Sensor Network and Parallel operating system.

\*\*\*\*\*

EC-7005(3)-CBGS